



Continuous Assessment Test 1 (CAT1) - August 2026

Programme	M.Tech.,	Semester	Fall 2026-27
Course	Machine Learning and Applications	Code	MACSE519
Faculty	Dr L Jeganathan	Slot/Class No.	C1+TC1/CH2026270103470
Time	90 Minutes	Max. Marks	50

Instructions:

- Answer all the questions.
- If any assumptions are required, assume the same and mention those assumptions in the answer script.
- Use of intelligence is highly appreciated.
- With your smartness, plan for a better time-management.
- In case of shortage of time, you can reduce the number of iterations to the level of at least two iterations.
- All type of calculators are allowed.

1. A company wants to predict the electricity consumption (r in kWh) of a machine based on its operating hours (x_1) and average load (x_2).

S.No	Operating Hours (x_1)	Average Load (x_2)	Energy Consumed (r)
1	2	15	32
2	3	18	39
3	5	20	49
4	6	24	58
5	8	28	73
6	10	30	84

Train a regression model (of your choice) to learn the relationship between x_1 , x_2 , and r , using two different hypotheses h_1 and h_2 , where h_1 is linear in parameters, involving all the input variables and h_2 has a term that involves both x_1, x_2 , in addition to all the terms of h_1 . Compute the training Mean Squared Error (MSE) for h_1 and h_2 . Based on the computed errors, identify which hypothesis learns the data better and justify your answer. If an iterative regression learning algorithm is used, do up to three iterations in all the calculations. [10 marks-CO2-3]

[Rubrics : Proposal of h_1, h_2 : 2 marks, Training both the models (with all the required calculations) : 4 marks, MSE computation : 2 marks, Decision on the better hypothesis (with reasons) : 2 marks.]

2. A company wants to predict the selling price (y in lakhs) of used cars based on their age (x in years).

S.No	Car Age (x)	Selling Price (y)
1	1	19.8
2	2	18.2
3	3	16.7
4	4	15.3
5	5	13.8
6	6	12.6

For the given dataset, choose any two regression hypotheses (of your choice) of increasing complexity. Estimate their parameters and arrange them from simplest to most complex. Identify which hypothesis underfits the data, which overfits the data. Based on your observation, suggest a hypothesis which may best fit the given dataset. For the hypothesis of best-fit, you are expected to provide justification without any mathematical calculations. [10 marks-CO2-3]

[Rubrics: Choice of the hypothesis : 3 marks, Calculation of parameters of h_1 : 2 marks, Calculation of parameters of h_2 : 2 marks, Justification for underfitting : 1 mark, Justification for overfitting : 1 mark, Hypothesis of best-fit : 1 mark.]

3. A real estate company wants to predict the selling price (y in lakhs) of houses using the following three features: x_1 (Built-up Area in sq.ft.), x_2 (Number of Bedrooms), x_3 (House Registration Number). The company trains two regression models, $M_1 : y = \theta_0 + \theta_1 x_1 + \theta_2 x_2$, $M_2 : y = \theta_0 + \theta_1 x_1 + \theta_2 x_2 + \theta_3 x_3$. The models are evaluated on a dataset containing $n = 20$ houses. The following statistics are obtained.

Model	R^2	RSS
M_1	0.89	220
M_2	0.90	218

Compute the Adjusted R^2 score for both models. Among the two models, identify which model performs well when trained with the same dataset. Identify the feature(s) of the dataset which is(are) relevant for the predicting the selling price with justification. Compute the MSE of both the models. Using the computed MSE values and the given R^2 values, comment on the relationship between prediction error and the coefficient of determination. [10 marks-CO2-2]

[Rubrics: Adjusted R^2 computation: 2 marks; Identification of better performing model: 2 marks; Identification of relevant feature(s): 2 marks; Computation of MSE: 2 marks. Relationship : 2 marks]

4. Consider the dataset $(x, y) = \{(5, 2), (2, -1), (3, 5), (4, 6)\}$ in a supervised learning setting with mean squared error loss. Propose two hypothesis classes, say A and B of different complexity, that could be used to model this data and estimate the parameters of A and B using the given samples. Based on the estimation of parameters, conclude which model is expected to exhibit higher bias and which is expected to exhibit higher variance. Justify your conclusion appropriately. [10 marks-CO2-3]

[Rubrics : Hypotheses A and B : 2 marks, Estimation of parameters of A and B : 5 marks, Conclusion on Bias and Variance : 3 marks]

5. Propose a problem (of your choice) related to the education sector that can be solved with a *regression learning model*. Clearly describe the proposed problem using the **E-T-P framework**. Explain E (with a minimum of three sample features), T , and P in detail, for the proposed problem. Further, describe the procedure for solving the problem using the regression learning model. You are not required to do any estimation of parameters for the learning model. **Note:** You should not describe a problem that was discussed in the classroom or provided in the practice problem sheets or laboratory problem sheets or the problems described in this question paper. [10 marks-CO1-3]

[Rubrics : Problem description : 2 marks, Description of a sample E : 3 marks, Description of the Task : 1 mark, Description of P : 2 marks, Description of the procedure : 2 marks]